Clustering and Classification with a Perceptual Distance Function

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Abstract

1 Introduction

Research in content-based image/video retrieval has steadily gained momentum in recent years as a result of the dramatic increase in the volume of digital images and videos. To achieve effective retrieval, an image/video system should be able to accurately characterize and quantify perceptual similarity. However, a fundamental challenge—how to measure perceptual similarity—remains largely unanswered.

Similarity is one of the most central theoretical constructs in psychology [4, 8]. It also plays a central role in information categorization and retrieval. Using our experimental results (for more detailed results, please refer to [??]), together with theories and examples in cognitive psychology, we discuss the progress of the following three similarity paradigms in cognitive psychology.

1. *Similarity is a measure of all respects.* Traditionally, Minkowski-type distance functions (particularly Euclidean distance) are widely used for measuring similarity. a Minkowski-like metric accounts for all respects (i.e., all features) when it is employed to measure similarity

between two objects. In [??], we mined feature distances (difference between individual features) between $60,000$ images and their similar images. Our mining result shown in Figure 1 is just one of a large number of counter-examples demonstrating that the assumption of the Minkowski-like metric is questionable. In Figure 1, the x-axis denotes the feature distance. The y-axis indicates the frequency of this feature distance occurs between similar images. As reveiled by the Figure, a moderate portion of feature distance is in the high value range (≥ 0.5), which indicates that similar images may be quite dissimilar in some features. From this observation, we infer that the assumption of the Minkowski metric is inaccurate. Similar images are not necessarily similar in all features (respects).

In addition, the psychology studies of [4, 8] also present examples showing that the Minkowski model appears to violate human similarity judgements.

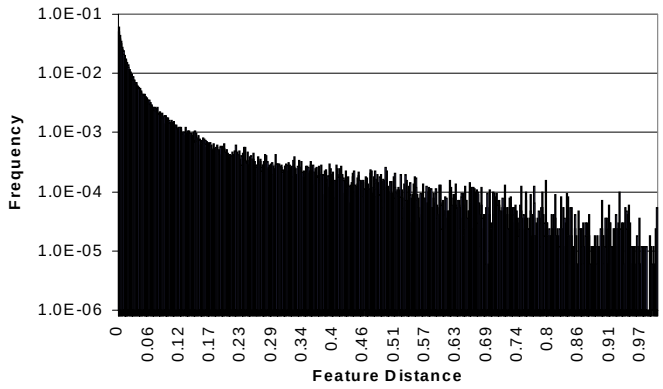


Figure 1: The Distribution of Feature Distances Between Similar Images.

2. *Similarity is a measure of a fixed set of respects.* Substantial work on similarity has been carried out by cognitive psychologists. The most influential work is perhaps that of Tversky [8], who suggests that

similarity is determined by matching features of compared objects, and integrating these features by the formula

$$S(A,B) = \theta f(A \cap B) - \alpha f(A - B) - \beta f(B - A). \quad (1)$$

The similarity of A to B, $S(A,B)$, is expressed as a linear combination of the common and distinct features. The term $(A \cap B)$ represents the common features of A and B. $(A - B)$ represents the features that A has but B does not; $(B - A)$ represents the features that B has but A does not. The terms θ, α, and β refect the weights given to the common and distinctive components, and function f is often assumed to be additive [4].

The weighted Minkowski function [7] and the quadratic-form distances [1, 3] are the two representative distance functions that match the spirit of Equation 1. The weights of the distance functions can be learned via techniques such as relevance feedback [6, 7], principal component analysis, and discriminative analysis [9]. Given some similar and some dissimilar objects, the weights can be adjusted so that similar objects can be better distinguished from other objects.

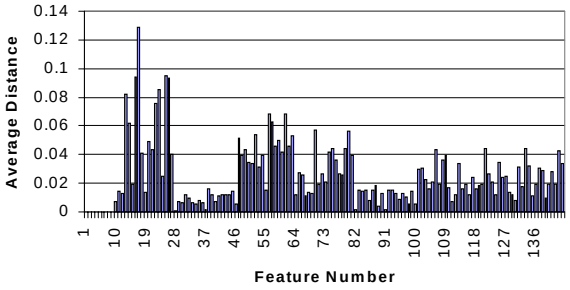
However, the assumption made by these distance functions, that all similar objects are similar in the same respects [9], is also questionable.We examined whether similar images resemble the query images in the same way in a $1.5M$ image dataset. We tallied the feature distance (denoted as δ^+) of the 144 features for different kinds of similar images. Figure 2 presents four representative similar images: GIF, cropped, rotated, and scaled images. The x-axis of the figure depicts the feature numbers, from 1 to 144. The first 108 features are various color features, and the last 36 are texture features. The figure shows that various similar images can resemble the query images in very different ways. GIF images have larger δ^+ in color features (the first 108 features) than in texture features (the last 36 features). In contrast, cropped images have larger δ^+ in texture features than in

color features. For rotated images, the δ^+ in colors comes close to zero, although its texture feature distance is much greater. A similar pattern appears in the scaled and the rotated images. However, the magnitude of the δ^+ of scaled images is very different from that of rotated images. It is apparent that similar images can be similar to a query image in differing respects.

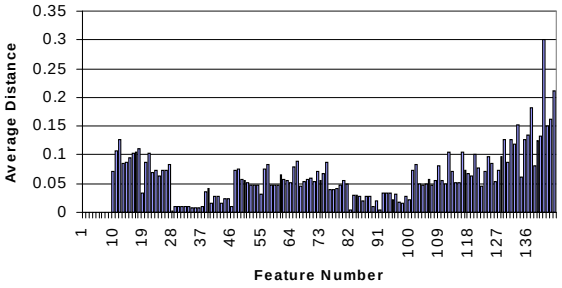
3. *Similarity is a process that provides respects for measuring similarity.* Suppose we are asked to name two places that are similar to England. Among several possibilities, Scotland and New England could be two reasonable answers. However, the respects England is similar to Scotland differ from those in which England is similar to New England. If we use the shared attributes of England and Scotland to compare England and New England, the latter pair might not be similar, and vice versa. This example echos the result presented in Figure 2—that is, objects can be similar to the query object in different respects. A distance function using a fixed set of respects cannot capture objects that are similar in different sets of respects.

   Murphy and Medin [5] provide early insights into how similarity works in human perception: "The explanatory work is on the level of determining which attributes will be selected, with similarity being at least as much a consequence as a cause of a concept coherence." Goldstone [2] explains that similarity is the process that determines the respects for measuring similarity. In other words, a distance function for measuring a pair of objects is formulated only after the objects are compared, not before the comparison is made. The respects for the comparison are activated in this formulation process. The activated respects are more likely to be those that can support coherence between the compared objects.

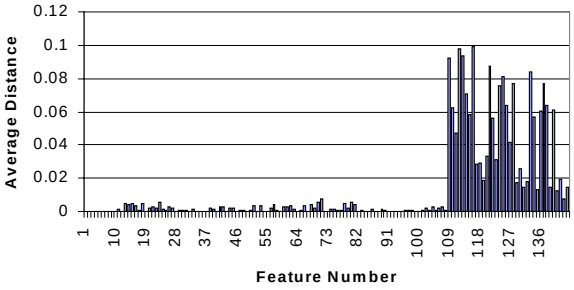
The above observations not only refute the assumptions of traditional distance functions, but also provide hints as to how a good distance func-



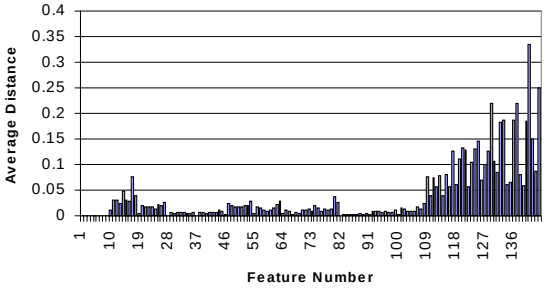
(a) GIF Images



(b) Cropped Images



(c) Rotated Images



(d) Scaled Images

Figure 2: The Average Feature Distances.

tion would work. The first point is that a distance function does not need to consider all features equally, since similar images may match only some features of the query images. The second point is that a distance function should weight features dynamically, since various similar images may resemble the query image in differing ways. These points lead to the design of the *dynamic partial* distance function.

Based on the observations explained above, we designed a distance function to better represent the perceptual similarity. Let $\delta_i = |x_i - y_i|$, for $i = 1, \cdots, p$. We first define sets Δ_m as

$$\Delta_m = \{The \quad smallest \quad m \; \delta's \quad of \quad (\delta_1, ..., \delta_p)\}.$$

Then we define the *Dynamic Partial Distance Function* (DPF) as

$$d(m, r) = (\sum_{\delta_i \in \Delta_m} {\delta_i}^r)^{\frac{1}{r}}. \tag{2}$$

DPF has two adjustable parameters: m and r. Parameter m can range from 1 to p. When $m = p$, it degenerates to the Minkowski metric. When $m < p$, it counts only the smallest m feature distances between two objects, and the influence of the $(p - m)$ largest feature distances is eliminated. Note that DPF dynamically selects features to be considered for different pairs of objects. This is achieved by the introduction of Δ_m, which changes dynamically for different pairs of objects. That is, DPF activates different features for different object pairs. The activated features are those with minimum differences–those which provide coherence between the objects. If coherence can be maintained (because sufficient a number of features are similar), then the objects paired are perceived as similar. In [??], we have shown that if a proper value of m is chosen, it is possible to make similar images aggregate more compactly and locate closer to the query images, simultaneously keeping the dissimilar images away from the query images. In other words, similar and dissimilar images can be better separated by DPF than by earlier methods.

As proven in ??, DPF is a non-metric distance function. Specifically, the triangle-inequality does not hold for DPF, Which makes the clustering and classification in DPF distance space much more chanllenging. In this study, we explored the clustering and classification methods in DPF space. The reminder of the paper is organized as follows: ...

2 Related Work

Clustering partitions objects into groups called clusters, so that similar objects fall into the same group. Data clustering has been extensively studied in the statistics, machine learning and pattern recognition. Lots of aglorithms [??] have been developed to cluster data in coordinate space, assuming Euclidean distance as the measure for similarity. However, as the distance function sways away from Euclidean distance function, the algorithms based on coordinates will yield deteriorated clustering quality.

A number of clustering algorithms have been proposed to cluster data in distance space, in which the only operation possible on data objects is the computation of distance. Among those, CLARANS is a representive one based on randomized search. CLARANS first find k medoids. It then uses these medoids as seeds to generate clusters. Of late, various methods [??] have been proposed to improve the I/O efficiency of CLARANS. Many of these modifications rely on vector operations, and thus limit their usage to coordinate space. Ganti et al. developed another distance-based clustering algorithm called BUBBLE and showed its scalability and efficiency in clustering large dataset. However, BUBBLE is designed on the foundation that the distance function satisfies the triangle inequality, which makes it impractical from clustering in non-metric distance space.

We selected CLARANS to cluster DPF for the following reasons:

1. CLARANS makes few assumptions about the working condition, which makes it robust enough for clustering non-metric distance functions.

2. CLARANS has been shown to be relatively efficient and effective in clustering large dataset.

3 Algorithm CLARANS

The CLARANS tackles the problem of partioning n points into k clusters. It is a medoid-based algorithm. That is, its objective is to find k best medoids that optimize a criterion funciton (usually the average distance to medoids). Each set of k medoids is represented as node in a graph, two nodes are adjacent if they have $k-1$ medoids in common. it starts with an arbitrary node. In each iteration, a random neighbor of the current node is tested to see whether it yields better cost. If this is the case, the neighbor is set as the current node. Iterations go on until some ending condition is satistified. The formal description of the algorithm is as follows:

Algorithm CLARANS

1. Input parameters *numlocal* and *maxneighbor*. Initialize i to 1, and *mincost* to a large number.

2. Set *current* to an arbitrary node in $G_{n,k}$.

3. Set j to 1.

4. Consider a random neighbor S of *current*, calculate the cost difference of the two nodes.

5. If S has a lower cost, set *current* to S, and go to Step (3).

6. Otherwise, increment j by 1. If $j \leq maxneighbor$, go to Step (4).

7. Otherwise, when $j > maxneighbor$, compare the cost of *current* with *mincost*. If the former is less than *minicost*, set *mincost* to the cost of *current*, and set *bestnode* to *current*.

8. Increment i by 1. If $i > numlocal$, output *bestnode* and halt. Otherwise, go to Step (2).

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